S. No.	Date	Title	Page No.	Teacher's Sign / Remarks

Need for Modulation :-

-> To transmit the message S/g for longer distance
-> To multiplex n number of message S/g and Transmit over a common channel
-> To reduce noise and interference
-> To reduce the height of Transmitting and receiving antennas for efficient transmission ~~are~~ and reception.

Principles of Communication System.

Preferred Book :-
Communication Systems - Simon Haykin
5th Edition.

Module (01) :- Amplitude Modulation :

+ **Communication System :-**
The purpose of the communication system is to transfer the information from source to destination VIA communi-
- cation channel.

+ **Source :-** It generates inform S/g.

+ Communication channel is of 2 types.
i) Wired ~~Channel~~ channel — Eg. Optical Fiber.
ii) Wireless Channel. — Eg : Radio channel.

+ **Types of Communication Systems :-**
i) Analog Communication System — Land Line System
ii) Digital Communication System — Wireless comm S/m

+ **Modulation :-**
The process of altering the parameter of carrier S/g in accorda-
- nce with the instantaneous values of message S/g keeping the other parameters constant

+ Based on this defn the

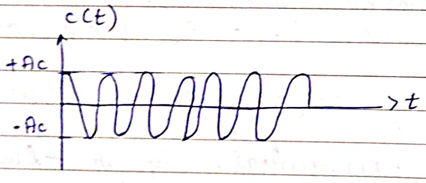
different characteristics of s/g are defined as follows.

i) Amplitude
ii) frequency.
iii) phase.

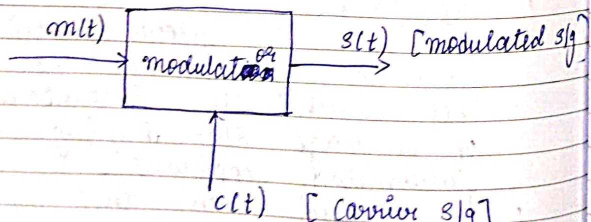
• Depending on these parameters, we have 3 modulation techniques
i) Amplitude Modulation
ii) Frequency Modulation
iii) Phase Modulation.

• Mathematically carrier s/g is defined by $c(t) = A_c \cos(2\pi f_c t + \phi_c)$

where
A_c = amplitude of carrier s/g in Volts
f_c = frequency of carrier s/g in Hz.
ϕ_c = phase of carrier s/g in radians



block diagram of Modulator :-



• Modulator is a device that operates with two I/P m(t) & c(t) and O/P s/g s(t)

m(t) is message s/g
(OR)
Information bearing s/g
(OR)
Baseband s/g.
(OR)
Modulating s/g.

• Amplitude ~~Modution~~ Modulation :-
The process of changing the amplitude of the carrier s/g with respect to ~~to~~ the message s/g by keeping the frequency and phase as constant.

8-10m • Time Domain Description of AM

• Let us consider the message s/g m(t) and a sinusoidal carrier s/g c(t). This carrier s/g is defined as
$c(t) = A_c \cos(2\pi f_c t)$
where c(t) is the instantaneous value of carrier s/g.
A_c = max amplitude of carrier s/g.
f_c = frequency of carrier s/g.

A/c to the def^n of AM the peak amplitude of a modulated s/g is given by $V_{AM} = A_c + m(t)$
Hence the s(t), modulated s/g is represented as $s(t) = V_{AM} \cos(2\pi f_c t)$
$= [A_c + m(t)] \cos(2\pi f_c t)$

$= A_c[1 + \frac{1}{A_c} m(t)] \cos(2\pi f_c t)$

where $\frac{1}{A_c} = K_a$ known as amplitude sensitivity of modulator which is responsible for generation of AM signal.

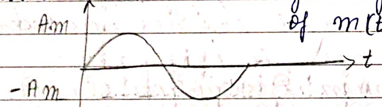
$$\boxed{s(t) = A_c [1 + K_a m(t)] \cos(2\pi f_c t)}$$

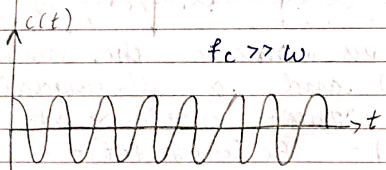
The above Eqn is called as ~~am~~ AM wave equation. where,

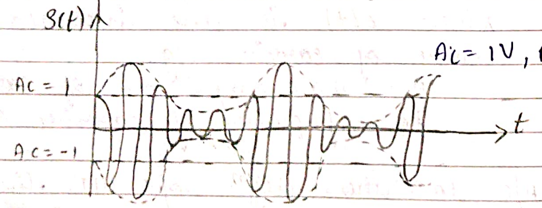
$[1 + K_a m(t)]$ = Envelope of AM wave

The values of $|K_a m(t)|$ results the following 3 cases.

- case (01) :- with $|K_a m(t)| < 1$ i.e., carrier s/g is under modulated. this condn is illustrated as follows. W = highest frequency component of m(t)

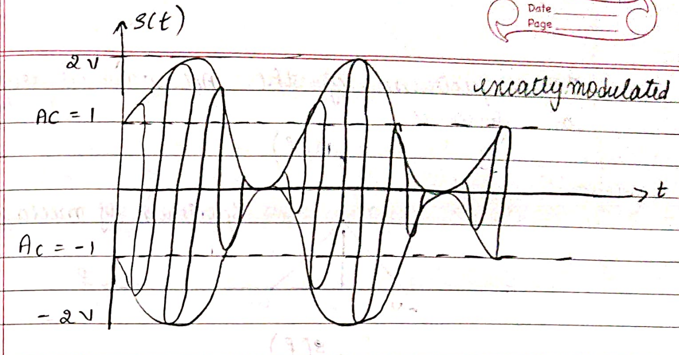






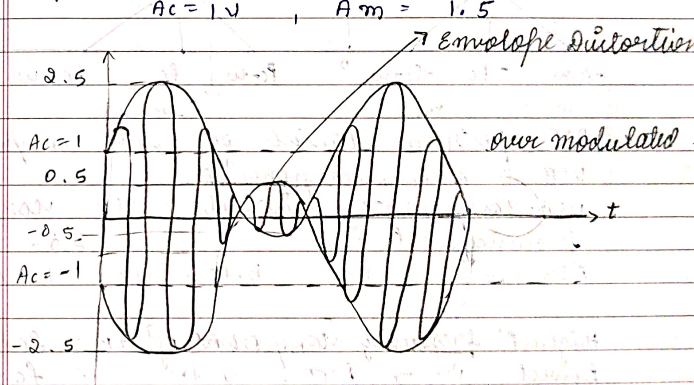
- case (02) :- with $|K_a m(t)| = 1$ i.e., carrier signal is exactly modulated

$A_c = 1V$, $A_m = 1V$



- case (03) :- with $|K_a m(t)| > 1$ i.e., carrier signal is overmodulated

$A_c = 1V$, $A_m = 1.5$



- Frequency Domain Description of AM wave

Consider Am modulated wave equation

$s(t) = A_c [1 + K_a m(t)] \cos[2\pi f_c t]$

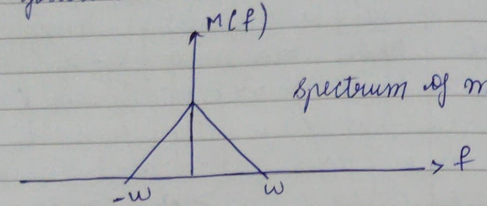
$= A_c \cos 2\pi f_c t + K_a A_c m(t) \cos 2\pi f_c t$

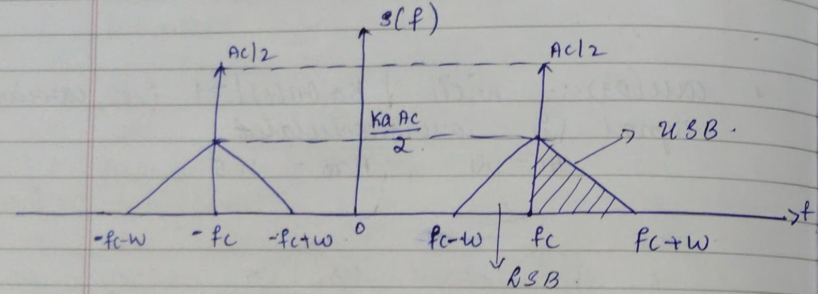
Taking fourier transform of a resulting equation we get

$S(f) = A_c\left[\frac{1}{2}[\delta(f - f_c) + \delta(f + f_c)]\right] + \frac{K_a A_c}{2}$

$[M(f - f_c) + M(f + f_c)]$

The spectrum of the AM wave is represented as follows





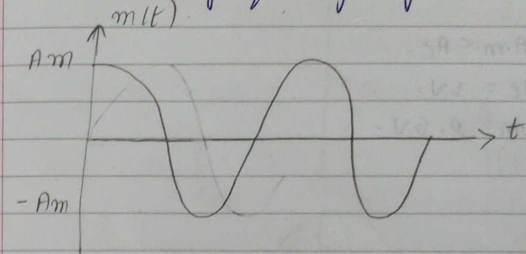
- AM spectrum consists of two components USB & LSB components.
USB components lies above the carrier frequency f_c.
LSB — — below — — —

Highest frequency component of AM = $f_c + W$
Lowest — — — = $f_c - W$
where W = bandwidth of message s/g
(or)
High frequency component of $m(t)$

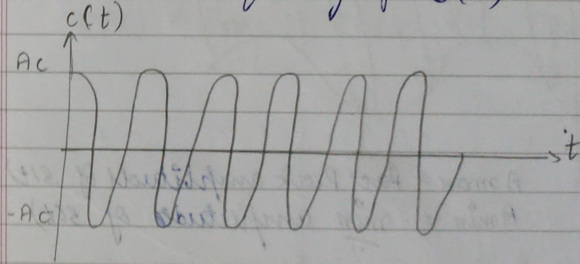
Band width of AM is given by = USB - LSB
= $\cancel{f_c} + W - (-\cancel{f_c} + W)$
= $\underline{2W}$

Single tone Modulation :-

Considering the modulating s/g $m(t)$ that consists of single tone ~~and~~ (or) single frequency component i.e.,
$m(t) = A_m \cos 2\pi f_m t$.
A_m = peak / max amplitude of $m(t)$
f_m = frequency of $m(t)$

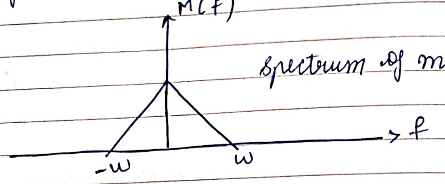


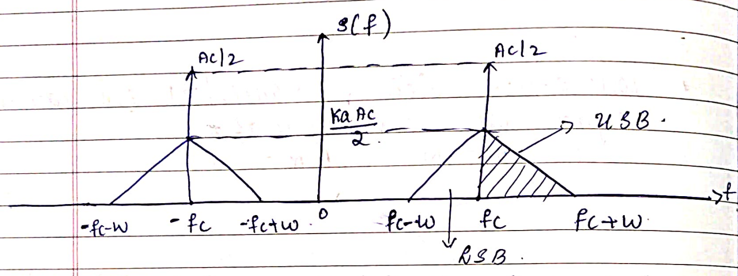
Let us consider sinusoidal carrier s/g.
$c(t) = A_c \cos 2\pi f_c t$.
A_c = Peak amplitude of $c(t)$.
f_c = Frequency of $c(t)$



Therefore the corresponding AM wave eq^n is given by.
$s(t) = A_c[1 + K_a m(t)] \cos 2\pi f_c t$
for single tone modulation $m(t) = A_m \cos 2\pi f_m t$
$\therefore$ for single tone modulated AM eq^n is $s(t) = A_c[1 + K_a A_m \cos 2\pi f_m t] \cos 2\pi f_c t$
$K_a A_m$ = modulation index
$\boxed{\frac{A_m}{A_c}} = \frac{\text{Peak amplitude of } m(t)}{\text{Peak amplitude of } c(t)}$

The spectrum of the AM wave is represented as follows





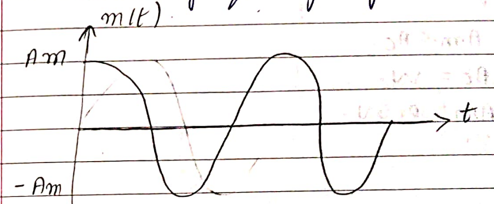
- AM spectrum consists of two components USB & LSB components.
  USB components lies above the carrier frequency. fc.
  LSB ,, ,, below ,, ,, ,,

Highest frequency component of AM = $fc+W$
Lowest ,, ,, ,, = $fc-W$
where W = bandwidth of message sig
(OR)
High frequency component of $m(t)$

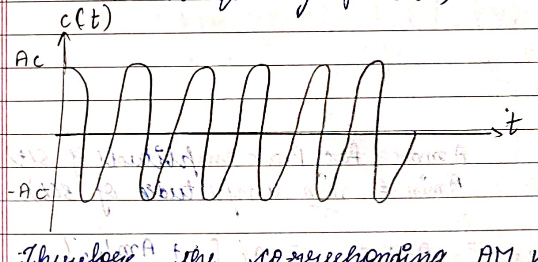
Band width of AM is given by = USB - LSB.
$= fc+W - fc+W$
$= 2W$

+ Single tone Modulation :-

Considering the modulating sig $m(t)$ that consists of single tone ~~and~~ (OR) single frequency component i.e.,
$m(t) = Am \cos 2\pi fm t.$
Am = peak / max amplitude of $m(t)$
fm = frequency of $m(t)$



But we consider sinusoidal carrier sig.
$c(t) = Ac \cos 2\pi fc t.$
Ac = Peak amplitude of $c(t)$.
fc = Frequency of $c(t)$



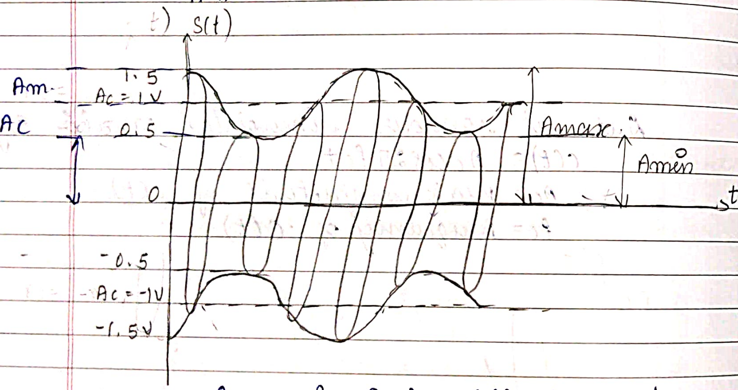
Therefore the corresponding AM wave eq^n is given by.
$s(t) = Ac[1+ Ka\, m(t)] \cos 2\pi fc t$
for single tone modulation $m(t) = Am \cos 2\pi fm t$
$\therefore$ for single tone modulated AM. eq^n is $s(t) = Ac[1+ Ka\, Am \cos 2\pi fm t] \cos 2\pi fc t$
$Ka\, Am$ = modulation index
$\frac{Am}{Ac}$ = Peak amplitude of $m(t)$ / Peak amplitude of $c(t)$

Note:-
To avoid envelope distortion μ must be kept below one.

$\mu < 1 \Rightarrow$ Under modulation.
$\mu = 1 \Rightarrow$ Exact modulation
$\mu > 1 \Rightarrow$ Over modulation.

+ Let us sketch single tone modulated s/g for $\mu < 1$

$A_m < A_c$.
$A_c = 1V$.
$A_m = 0.5V$.



A_{max} = ~~fax~~ Peak amplitude of $s(t)$
A_{min} = min amplitudes of $s(t)$

$$\frac{A_{max}}{A_{min}} = \frac{A_c + A_m}{A_c - A_m} = \frac{A_c[1 + A_m/A_c]}{A_c[1 - A_m/A_c]}$$

$$\boxed{\frac{A_{max}}{A_{min}} = \frac{1+\mu}{1-\mu}}$$

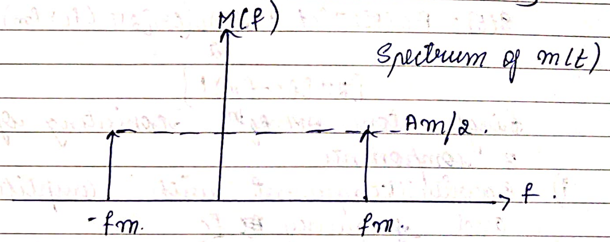
$$A_{max} - \mu A_{max} = A_{min} + \mu A_{min}$$
$$A_{max} - A_{min} = \mu(A_{min} + A_{max})$$
$$\boxed{\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}}$$

+ **Spectrum of message s/g m(t):-**

$m(t) = A_m \cos 2\pi f_m t$.

By taking Fourier transform of $m(t)$ the spectrum of message s/g is given by

$$M(f) = \frac{A_m}{2}\left[\delta(f - f_m) + \delta(f + f_m)\right]$$

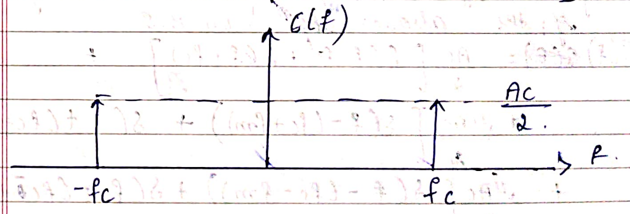


+ **Spectrum of carrier s/g c(t)**

$c(t) = A_c \cos 2\pi f_c t$

Taking F.T of $c(t)$, the spectrum of message s/g is given by

$$C(f) = \frac{A_c}{2}\left[\delta(f - f_c) + \delta(f + f_c)\right]$$



+ **Spectrum of amplitude modulated s/g**

Let us consider single tone modulated s/g.

$$s(t) = A_c[1 + \mu \cos 2\pi f_m t]\cos 2\pi f_c t$$
$$s(t) = A_c \cos 2\pi f_c t + \mu A_c \cos 2\pi f_c t \cos 2\pi f_m t$$
$$= A_c \cos 2\pi f_c t + \frac{\mu A_c}{2}\left[\cos(\omega_c + \omega_m) + \cos(\omega_c - \omega_m)\right.$$

Using trignometric func $\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$

the above Eqn can be simplified as follows.

$$s(t) = A_c \cos 2\pi f_c t + \frac{\mu A_c}{2}[\cos[2\pi(f_c+f_m)t] + \frac{\mu A_c}{2}[2\pi[f_c - f_m)t]$$

Single tone AM Eqn consisting of following 3 components

i) carrier component with amplitude A_c and frequency f_c

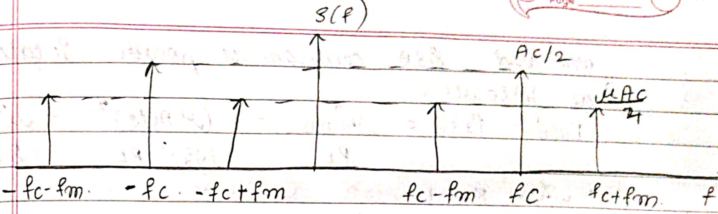
ii) USB component with amplitude $\mu A_c/2$ and frequency (f_c+f_m)

iii) LSB component with amplitude $\mu A_c/2$ and frequency $(f_c - f_m)$

- Spectrum of single tone AM is given by taking fourier transform of the above Eqn on B.S.

$$S(f) = \frac{A_c}{2}[\delta(f-f_c)+\delta(f+f_c)] + \frac{\mu A_c}{4}[\delta(f-(f_c+f_m)) + \delta(f+(f_c+f_m))] + \frac{\mu A_c}{4}[\delta(f-(f_c-f_m)) + \delta(f+(f_c-f_m))]$$

The spectrum of Am s/g is given by.



The highest frequency component $= f_c + f_m$

The lowest frequency component $= f_c - f_m$

Bandwidth $= HFC - LFC$

$= f_c + f_m - f_c + f_m$

$= \underline{2f_m}$

5M

Total Power of AM wave

Consider AM wave Eqn for single tone

$$s(t) = A_c \cos 2\pi f_c t + \frac{\mu A_c}{2}[\cos(2\pi(f_c+f_m)t] + \frac{\mu A_c}{2}[\cos 2\pi(f_c-f_m)t]$$

Since this Eqn consisting of three components such as carrier, USB, LSB the total power of AM wave is given by

$$\boxed{P_t = P_c + P_{USB} + P_{LSB}}$$

where

P_c = carrier power.

$c(t) = A_c \cos 2\pi f_c t$

$P_c = \frac{V_{rms}^2}{R_L}$ where $V_{rms} = A_c/\sqrt{2}$

$P_c = \frac{(A_c/\sqrt{2})^2}{R_L} = \frac{A_c^2}{2R_L}$.

with $R_L = 1\Omega$,

$P_c = \frac{A_c^2}{2}$.

USB and LSB components power is calculated as follows.

$$P_{USB} = P_{LSB} = \frac{V_{rms}^2}{R_L} = \frac{(\mu A_c/2)^2}{(\sqrt{2})^2 R_L} = \frac{\mu^2 A_c^2}{8 R_L}$$

with $R_L = 1$,

$$P_{USB} = P_{LSB} = \frac{\mu^2 A_c^2}{8}$$

$$\therefore \; P_t = \frac{A_c^2}{2} + \frac{\mu^2 A_c^2}{8} + \frac{\mu^2 A_c^2}{8}$$

$$= \frac{A_c^2}{2} + \frac{\mu^2 A_c^2}{4}$$

$$= \frac{A_c^2}{2}\left[1 + \frac{\mu^2}{2}\right]$$

$$\boxed{P_t = P_c\left[1 + \frac{\mu^2}{2}\right]}$$

- Efficiency of AM wave :-

$$\eta = \frac{\text{SB Power}}{\text{Total Power of AM wave}}$$

$$= \frac{P_{USB} + P_{LSB}}{P_T}$$

$$= \frac{\frac{\mu^2 A_c^2}{4}}{P_c\left[1 + \frac{\mu^2}{2}\right]} = \frac{\frac{\mu^2}{2} \times \cancel{P_c}}{\cancel{P_c}\left[1 + \frac{\mu^2}{2}\right]}$$

$$= \frac{\frac{\mu^2}{2}}{\left(1 + \frac{\mu^2}{2}\right)} = \frac{\frac{\mu^2}{\cancel{2}}}{\frac{2 + \mu^2}{\cancel{2}}}$$

$$= \frac{\mu^2}{2 + \mu^2}$$

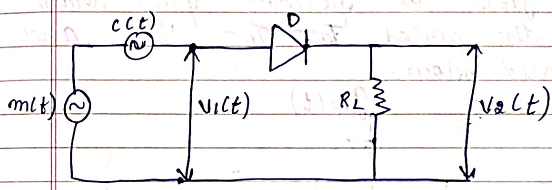
- AM Modulator :-

It is used to generate amplitude modulated S/g. The generation AM S/g may be accomplished using various devices.

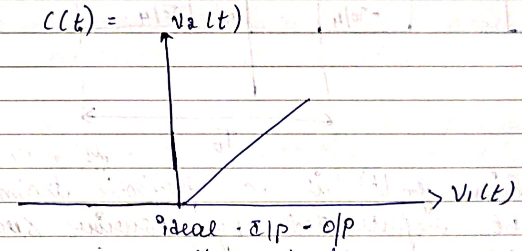
① Switching modulator
② Square law modulator

① Switching ~~Diode~~ Modulator :-

The ckt diagram of switching modulator is shown below



Assume



ideal I/P - O/P characteristics

D - ideal diode

- $|m(t)| \ll c(t)$

The I/P S/g $v_1(t)$ is given by

$$v_1(t) = m(t) + c(t)$$

$$= m(t) + A_c \cos 2\pi f_c t$$

Since $|m(t)| \ll A_c$, the load voltage $v_2(t)$ is given by $v_2(t) \simeq v_1(t)$ when $c(t) > 0$

• when $c(t) > 0$, the diode conducts and function as a short ckt

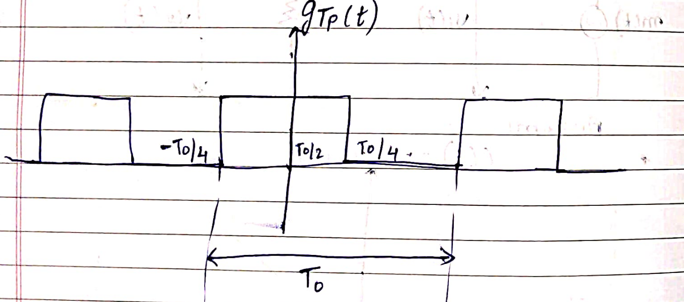
$$V_2(t) = \begin{cases} V_1(t), & c(t) > 0 \\ 0, & c(t) < 0 \end{cases}$$

i.e., the load voltage $V_2(t)$ varies periodically b/w the values $V_1(t)$ and zero at a rate = frequency of carrier s/g.

• The load voltage $V_2(t)$ may be defined as $V_2(t) \simeq V_1(t)\, g_{T_P}(t)$

$$\simeq [m(t) + A_c \cos 2\pi f_c t]\, g_{T_P}(t)$$

where $g_{T_P}(t)$ is periodic pulse train with the period $T_0 = 1/f_c$ and illustrated below.



Since $g_{T_0}(t)$ is a periodic train, it can be represented by Fourier series as follows

$$g_{T_0}(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)} \cos[2\pi f_c t (2n-1)]$$

by substituting this eqn in $V_2(t)$, we get

$$V_2(t) \simeq [m(t) + A_c \cos 2\pi f_c t]\left[\frac{1}{2} + \frac{2}{\pi}\cos 2\pi f_c t + HFC\right]$$

$$= \frac{1}{2} m(t) + \frac{A_c \cos 2\pi f_c t}{2} + \frac{2\, m(t)}{\pi} \cos 2\pi f + HFC$$

$$= \frac{A_c}{2}\left[1 + \frac{4\, m(t)}{\pi A_c}\right] \cos 2\pi f_c t$$

In this case $K_a = \frac{4}{\pi A_c}$

• All unwanted components are eliminated by passing $V_2(t)$ through band pass filter with the center frequency f_c and bandwidth $2W$ Hz

• Amplitude Demodulator :-

Demodulation :- It is the process of recovering the message from modulated signal $s(t)$.

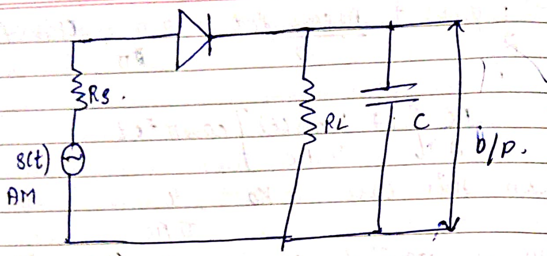
AM Demodulator :- Device used to recover the original message s/g from AM modulated s/g.

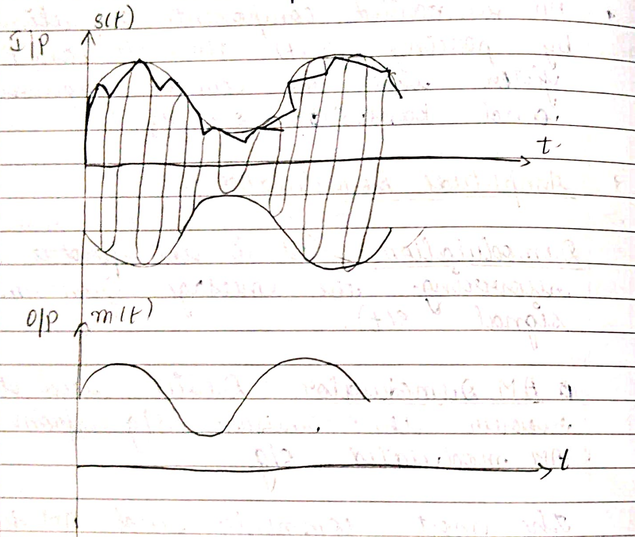
The most commonly used AM detectors are square law detector and envelope detector

Envelope Detector :- It is a device used to recover the original $m(t)$ from the AM modulated s/g.

→ The func of the envelope detector is the one which detects the envelope of the $m(t)$.

→ The ckt diagram of envelope detector is shown below





+ Envelope detector consists of diode followed by RC filter

Opr^n :-

-> During +ve half cycle of a I/P S/g the diode is forward biased and the capacitor charges towards peak value of a I/P S/g

-> When the I/P S/g falls below this ~~vatte~~ value ~~tha~~ the diode value becomes reverse biased and the capacitor C discharges through RL

-> As a result only +ve half cycle of a AM ~~S/g~~ wave appears across RL

-> This discharging process continues untill the next +ve half cycle.

-> When the I/P S/g becomes greater than the voltage across the capacitor the diode gets forward biased and starts conducts again and process is repeated.

-> The capacitor charges through source resistance Rs hence the charging time constant is given by

$R_s \times C << 1/f_m$

-> The capacitor discharge through RL hence the discharge time const is given by $R_L \times C << 1/\omega$

-> For proper demodulation ~~1/fc~~

$\boxed{1/f_c << R_L << 1/f_m}$

+ Advantages, Disadvantages and Modified form of AM :-

Advantages :-

-> The ckt of AM modulator and AM demodulator is very simple.

-> Generation of AM wave and demodulated wave ~~is~~ ~~eas~~ becomes easier

-> AM modulated wave can be travelled for the long distance.

Disadvantage :-

-> Waste of ~~carrying~~ carrier power.

-> Requires more bandwidth.

+ Modified form of AM :-

To overcome the limitations from the std AM wave the following 3 ~~techniques~~ modified form of AM wave is considered.

i) Double side band Suppressed carrier - DSBSC

$S(t) = USB + LSB$

$2W$

$P(t) = P_{USB} + P_{LSB}$

ii) Single side band. - SSB -

$S(t) = USB$ or LSB

$P(t) = P_{USB}$ or P_{LSB}

iii) Vestigial side band - VSB.

8m + Double Side band Suppressed carrier :-

DSBSC is a method of transmitting two side bands [LSB & USB] supressing carrier s/g. Therefore this method overcomes the drawback of power wastage in std AM.

Time Domain Description Of DSBSC :-

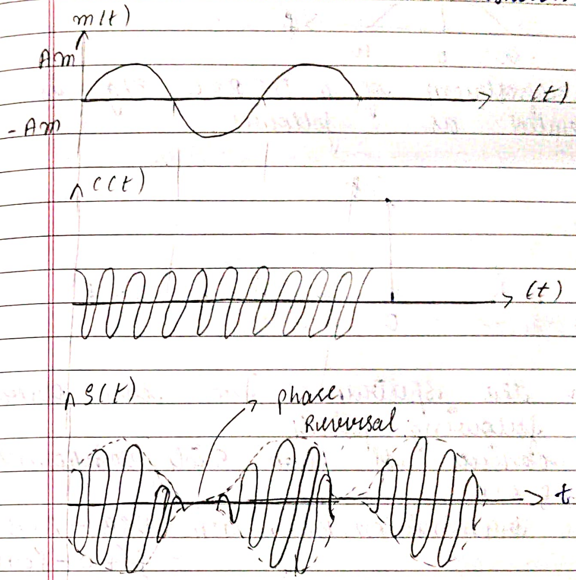
DSBSC waveform can be generated by multiplying the carrier s/g $c(t)$ with message s/g $m(t)$

Let $m(t) \Rightarrow$ message s/g with the frequency W Hz. and carrier s/g $c(t) = A_c \cos 2\pi f_c t$. Hence the DSBSC modulated s/g is given by. -

$s(t) = c(t) \cdot m(t)$

$s(t) = A_c \cos 2\pi f_c t \cdot m(t)$

The I/P and O/P waveforms of DSBSC modulator are shown below.



→ The DSBSC modulated s/g undergoes phase reversal ~~at~~ whenever the $m(t)$ crosses zero.

+ Frequency Domain description :-

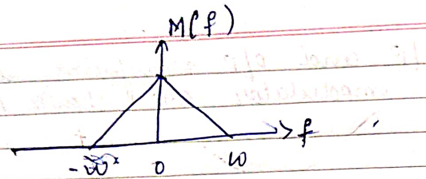
Consider the DSBSC wave Eq^n.

$s(t) = A_c \cos 2\pi f_c \, m(t)$

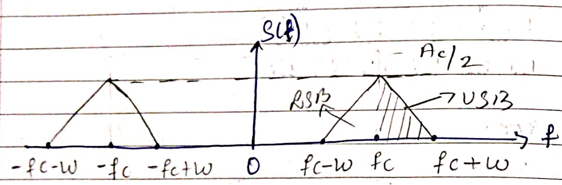
Taking Fourier Transform of above on B.S. we get.

$$S(f) = \frac{A_c}{2}\left[M(f-f_c) + M(f+f_c)\right]$$

The spectrum of the message s/g is illustrated as follows.



The spectrum of a DSBSC S/g is illustrated as follows.



From the spectrum we can observe the following points.

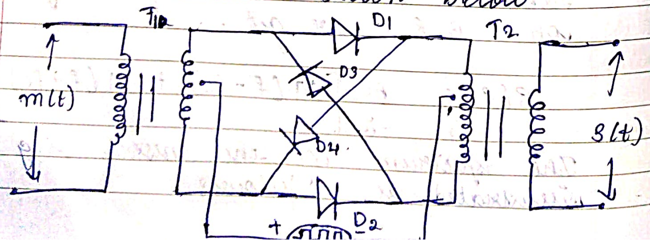
→ It consists of two side components USB & LSB.

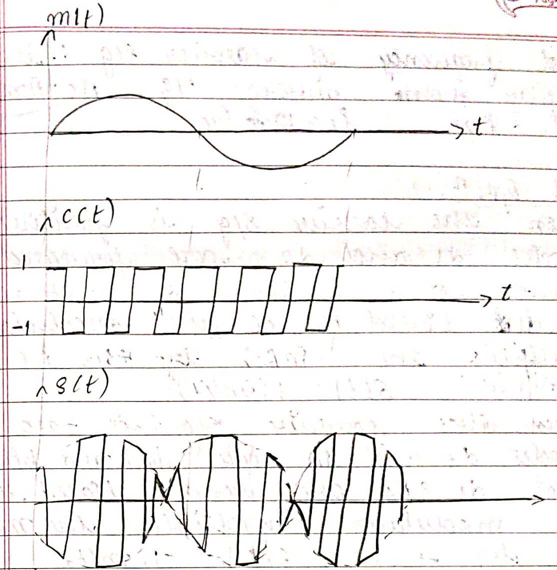
→ The transmission Bandwidth = USB − LSB
= $f_c+W - f_c+W$
= $2W$

which is same as Std AM. and there is no delta fun^c.

→ Generation of DSBSC wave :-

→ Ring Modulator :-
The ckt diagram of a Ring modulator is shown below.





Ring Modulator is a ~~...~~ one ~~in~~ which is used to generate DSBSC wave

→ It consists of 1:1 transformers T1 and T2 and the 4 diodes d1, d2, d3, d4 in the Bridge form.

→ Operation :-

→ When c(t) is 1, ~~the/value~~ the diodes d1 & d2 are forward Biased. So the s(t) becomes $1 \times m(t) = m(t)$

→ When c(t) is −1, the diodes d3 & d4 are forward Biased. So the s(t) becomes $-1 \times m(t) = -m(t)$.

→ Ring modulator is a product modulator used to generate DSBSC s/g

→ ~~For work~~ It is assumed that amplitude and frequency of carrier s/g $\therefore$ is greater than message s/g $A_c > A_m$ & $f_c > f_m$, $f_c = 10 f_m$.

Ckt Oper^n:-

$\rightarrow$ When the carrier s/g is positive the diodes d1 and d2 are forward biased and diodes d3 & d4 are reverse biased. Hence the modulator multiplies the $m(t)$ by +1 i.e., $s(t) = 1 . m(t)$

$\rightarrow$ When the carrier s/g is -ve the diodes d3 and d4 are forward biased and d1 & d2 are reverse biased. Hence the modulator multiplies the message s/g by -1. i.e., $s(t) = -1 . m(t)$.

$\rightarrow$ The square wave carrier ~~c(t)~~ s/g can be represented by Fourier series as follows.

$$c(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)} \cos\left[2\pi f_c t (2n-1)\right]$$

$$= \frac{4}{\pi}\cos 2\pi f_c t - \frac{4}{3\pi}\cos(6\pi f_c t) + HFC. \quad \text{---} (1)$$

$\rightarrow$ The ring modulator o/p is given by

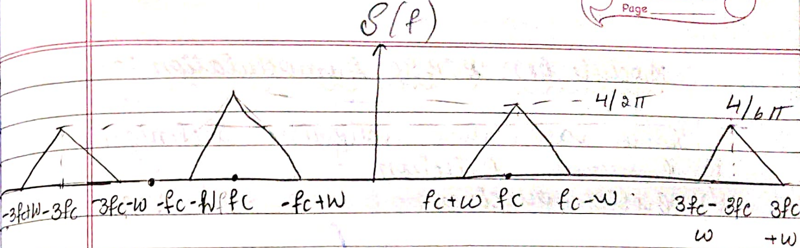
$$s(t) = c(t) . m(t)$$

$$s(t) = \left[\frac{4}{\pi}\cos 2\pi f_c t - \frac{4}{3\pi}\cos(6\pi f_c t) + HFC\right] . m(t) \quad \text{---} (2)$$

$\cos 2\pi(3 f_c t)$

After taking fourier tranform of a resulting s/g ~~we~~ get on B.S we get.

$$S(f) = \frac{4}{2\pi}\left[M(f-f_c) + M(f+f_c)\right] - \frac{4}{6\pi}\left[M(f-3f_c) + M(f+3f_c)\right] + HFC$$



$\rightarrow$ The modulated waveform is obtained by ~~passing~~ passing $s(t)$ through a band pass filter with the center frequency f_c and bandwidth $2W$. Hence final modulated s/g is given by.

$$\hat{} \; \frac{4}{\pi} m(t)\cos 2\pi f_c t$$

(1) A 500W carrier is modulated on a depth of 70%. Calculate the total power in the modulated wave in the following forms of a AM.

i) DSBSC with full carrier

ii) DSBSC ~~w~~ suppressed carrier

iii) Single Side band suppressed carrier.

$\Rightarrow$ Given:-

$P_c = 500W$.

$\% \mu = 70\% \Rightarrow 0.7$.

i) $P_t = P_c\left[1 + \frac{\mu^2}{2}\right]$

$= 500\left[1 + \frac{0.7}{2}\right]$

$P_t = 675W$

ii) $P_t = \frac{P_c \mu^2}{2} = \frac{500 \times (0.7)^2}{2} = 122.5W$

iii) $P_t = \frac{P_c \mu^2}{2} \times \frac{1}{2} = \frac{122.5}{2} = 61.25W$